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Facultatea Calculatoare, Informatică şi Microelectronică

Laboratory Work No. 2:

Monoalphabetic Cipher Cryptanalysis

Variant V15

Course: Cryptography and Security

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1 SHORT THEORY

The objective of this laboratory work is to recover the original English text from a message encrypted with a monoalphabetic substitution cipher. In this cipher, every plaintext letter is consistently replaced with one ciphertext letter. Letter positions and repetition patterns are preserved, while the identities of the letters are concealed.

A sufficiently long ciphertext retains statistical features of the language in which its plaintext was written. English letters such as E, T, A, O, I, and N are common, while others occur much less frequently. Frequency ranking gives initial hypotheses for the substitution, but is not sufficient by itself. Short words, repeated letter patterns, digraphs, trigraphs, and the meaning of the emerging text must also be checked.

2 CRYPTANALYSIS METHODOLOGY

2.1 Input and counting method

The analyzed message is Variant 15 (V15) from the laboratory handout. The frequency count includes alphabetic characters only, ignoring case; spaces, punctuation, digits, and hyphens are not counted. A total of 3165 letters were counted. The most frequent ciphertext symbol is `V`, appearing 438 times (13.84%).

2.2 Letter-frequency distribution

The ciphertext letters were counted and sorted in descending order of frequency. Table 1 shows the count and the percentage of the total for every ciphertext letter present in the message.

Table 1: Letter frequencies in ciphertext V15.

Cipher	Count	Frequency (%)	Cipher	Count	Frequency (%)
V	438	13.84	U	91	2.88
W	283	8.94	D	71	2.24
T	243	7.68	J	65	2.05
N	227	7.17	C	63	1.99
I	222	7.01	F	62	1.96
P	212	6.70	R	51	1.61
G	207	6.54	A	38	1.20
X	188	5.94	K	21	0.66
Q	159	5.02	L	16	0.51
O	144	4.55	Y	9	0.28
H	127	4.01	B	4	0.13
S	118	3.73	M	4	0.13
Z	100	3.16	E	2	0.06

The most frequent symbol, `V`, is a strong candidate for plaintext E. The next most frequent symbols provide further candidates, but individual letter rankings in one passage can differ from the average English ranking. For that reason, the hypotheses were verified using recognizable word patterns rather than accepted from frequency alone.

2.3 Progressive substitutions

The repeated trigraph `WQV` appears in several positions and is consistent with the common English word "the". This gives the assignments `W`→T, `Q`→H, and `V`→E. The result agrees with the high frequency of `V`.

The ciphertext word `WVSVJITUQ` then decrypts to "TELEGRAPH", and `hifuwnjituqf` decrypts to "cryptography". These words establish or confirm additional substitutions, including `S`→L, `J`→G, `I`→R, `T`→A, and `U`→P. The phrase `ztov hifuwnjituqf` becomes "made cryptography" and provides an independent contextual check.

Further patterns produce readable phrases such as "Samuel F. B. Morse sent 'What hath God wrought!' in 1844" and "The Secret Corresponding Vocabulary". The proposed substitutions were applied consistently across the entire cryptogram. Word boundaries merged in the supplied text layout were restored only where the surrounding text made the separation unambiguous; this does not change the letter-frequency counts or substitution key.

2.4 Reconstructed substitution alphabet

The recovered correspondence is shown in Table 2. The top row of each block lists ciphertext letters, and the lower row gives their plaintext equivalents.

Table 2: Recovered substitution mapping for V15 (ciphertext to plaintext).

A	B	C	D	E	F	G	H	I	J	K	L	M
B	Q	F	U	J	Y	N	C	R	G	V	K	Z
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
O	D	S	H	W	L	A	P	E	T	I	X	M

The mapping is a one-to-one substitution over the English alphabet. It recovers the letters of the passage consistently; the few absent spaces in the supplied text are a transcription/layout issue and are not part of the letter mapping.

3 RESULTS

The recovered plaintext describes how the telegraph affected commercial, governmental, military, and diplomatic cryptography. The text below is the decrypted Variant 15 passage. A small number of word boundaries were restored where the provided ciphertext transcription joined adjacent words.

> THE TELEGRAPH made cryptography what it is today. Samuel F. B. Morse sent "What hath God wrought!" in 1844. The next year his lawyer and promotional agent, Francis O. J. Smith, published a commercial code entitled The Secret Corresponding Vocabulary; Adapted for Use to Morse's Electro-Magnetic Telegraph, in whose preface he declared that "secrecy in correspondence, is far the most important consideration." This was provided by a super-encipherment.
>
> As the most exciting invention of the first half of the century, the telegraph stirred as much interest in its day as Sputnik did in its. The great and widely felt need for secrecy awakened the latent interest in ciphers that so many people seem to have, and kindled a new interest in many others. Dozens of persons attempted to dream up their own unbreakable ciphers. Their contributions enriched it with dozens of new cipher systems.

As businessmen and the public used the telegraph more and more, they found that their fears about lack of privacy were exaggerated. The clerks dealt impersonally with the messages. The telegraph companies respected their confidentiality. And commercial codes like Smith's, which replaced words and phrases by single codewords or code-numbers to cut telegraph tolls, afforded sufficient security for most business transactions by simply precluding an at-sight comprehension of the meaning. The brokers and traders soon realized that the main advantage of these codes was their economy.

Government ministries used the telegraph, too. At first they must have encoded with their nomenclators. But although secrecy was paramount for them, they liked the telegraphic economy of a large code—especially as they telegraphed more and more. So when the time arrived to compile a new nomenclator, they abandoned that form, copied the commercial form, and produced a full-fledged code. The nomenclators had had their 1,- or 2,000 code-numbers in mixed order, but the war and foreign ministries balked at the expense of drawing up a 50,000-entry code in two parts, and they had no professional cryptanalysts to warn them of the danger of the one-part format. They relied for security upon small editions, big safes, extensive lexicon (large codes are harder to break than small ones, other things being equal), and superencipherment, retaining codenumbers to facilitate this instead of switching to codewords. This evolution was essentially complete by the 1860s. The large, one-part code had replaced the small, two-part nomenclator in high-level military and diplomatic cryptography.

Meanwhile, the telegraph, author of this development, was creating something new in war—signal communications, or voluminous command and reconnaissance messages. Of course such messages had existed before, with torches, pigeons, and couriers, but in so rarefied a form that they were not even called "signal communications." The telegraph enabled commanders, for the first time in history, to exert instantaneous and continuous control over great masses of men spread over large areas.

These tactical messages required protection: telegraph wires could be tapped. Neither the old nomenclator nor the new code would do. They were too easy to capture in combat, too hard to reissue quickly and frequently to the numerous and widespread telegraph posts. Signal officers turned away from them. They looked instead to that neglected child of cryptography, the cipher. Ciphers could be printed cheaply on a single sheet of paper and distributed with ease. Secrecy was based upon variable keys, so capture of the general system and even of one of the keys would not compromise all an army's secret messages. Solutions would be prevented by rapid key changes. Ciphers were ideal for battle-zone messages, and the first of the modern wars, the American Civil War, used them for just that. Thus was born a new genre in cryptography: the field cipher.

4 CONCLUSIONS

The frequency analysis of 3,165 letters provided a useful starting point for cryptanalysis. The high frequency of ciphertext `V` suggested plaintext E, while the repeated trigraph `WQV` identified “the” and fixed three important substitutions. Further recognizable words, especially “telegraph” and “cryptography”, supported the remaining letter mappings.

The substitution key recovered every letter consistently. The result illustrates the weakness of monoalphabetic substitution for a sufficiently long message in a known language: its frequency distribution and preserved word patterns reveal the underlying plaintext without trying every possible permutation of the alphabet.